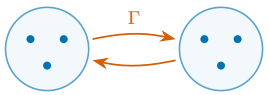
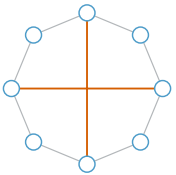
a **The same individual rules define two communities and finite meta-networks**



$M = 2$; $N = n_1 + n_2$

$$B_\Gamma = \begin{pmatrix} 1-\Gamma & \Gamma \\ \Gamma & 1-\Gamma \end{pmatrix}$$

same local update rules



Finite number of communities M

$$N = \sum_{c=1}^{M} n_c \text{ total agents}$$

M: number of communities

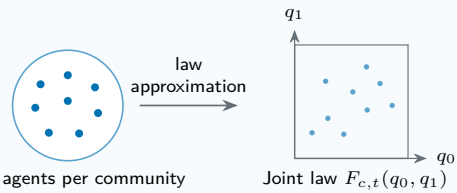
n_c: agents in community c

Equal sizes give $N = Mn$.

$n_{\min} \to \infty$; fixed M

$M \to \infty$; fixed n

b **Large communities: a network of joint score laws**

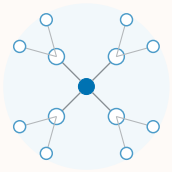


Many agents per community

Joint law $F_{c,t}(q_0, q_1)$

Individual outcomes remain random.
Community laws evolve deterministically.

c **Many communities: a stochastic rooted-graph process**



Community blocks retain finite-agent randomness.

$\theta = 0$: local exposure only.

Marked local graph convergence is required.

Root and local dependency cone

Empirical averages additionally require a two-root convergence condition.

Both scales grow $M \to \infty$, $n_{\min} \to \infty$; $\theta = 0$

A nonlinear tagged agent evolves in a network of joint score laws.
The marked sparse-graph assumptions remain necessary.

Every population-limit statement fixes the time horizon T. Increasing N alone does not specify a limit.

Finite-time convergence does not select a unique long-run state. Schematics do not represent sampled data.